Background: On-Policy Distillation

- Student model π_θ: Samples trajectories on-policy from its own strategy.
- Teacher model π_T: Provides per-token scores on the student's trajectories.
- Loss Function: Minimize the **reverse KL** divergence from student to teacher.

$$D_{KL}(\pi_\theta \| \pi_T) = \mathbb{E}_{x \sim \pi_\theta}\Big[\log \pi_\theta(x_{t+1}|x_{1..t}) - \log \pi_T(x_{t+1}|x_{1..t})\Big].$$

- Core Objective: To make the student's behavior mimic the high-performance teacher on states the student actually visits.

Limitation 1: Student No Better than the Teacher

- The training signal is entirely defined by the teacher's distribution:
  - The reverse KL loss directly penalizes deviation from the teacher.
  - The optimal solution is $\pi_\theta \approx \pi_T$ (limited by student capacity).

Limitation 2: Mode-Seeking and Reduced Diversity

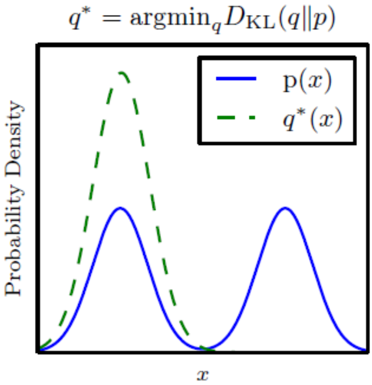

Figure: Reverse KL

Goals of This Section

Two questions:

1. Where does $\pi^*(y \mid x) \propto \pi_T(y \mid x)^\lambda \exp(\beta r(x, y))$ come from?
2. What is the exact sequence-level Bayesian distillation algorithm?

We answer both using standard Bayes' rule and importance weighting.

From Standard Bayes to Target Policy

Introduce a binary outcome:

$$o \in \{0, 1\}, \quad o = 1 \text{ means "good / correct"}.$$

Bayes' rule:

$$p(y \mid x, o = 1) \propto p(y \mid x)\, p(o = 1 \mid x, y).$$

Model choices:

$$p(y \mid x) \propto \pi_T(y \mid x)^\lambda$$
$$p(o = 1 \mid x, y) \propto \exp\left(\beta r(x, y)\right)$$

Then:

$$\pi^*(y \mid x) := p(y \mid x, o = 1) \propto \pi_T(y \mid x)^\lambda \exp\left(\beta r(x, y)\right).$$

This is the unified target policy.

Learning the Outcome Model from 0/1 Labels

Assume a labeled dataset:

$$\mathcal{D} = \{(x_i, y_i, r_i)\}, \quad r_i \in \{0, 1\}.$$

Define a parametric model:

$$p_\phi(o = 1 \mid x, y) = \sigma\big(g_\phi(x, y)\big),$$

train ϕ with binary cross-entropy:

$$L_{\text{out}}(\phi) = -\mathbb{E}_{(x,y,r)\sim\mathcal{D}}\big[r \log p_\phi(o = 1 \mid x, y) + (1-r) \log(1 - p_\phi(o = 1 \mid x,$$

We set the score:

$$r(x, y) := g_\phi(x, y),$$

Posterior Target Policy (Explicit Form)

With the learned outcome model:

$$\pi^*(y \mid x) = \frac{1}{Z(x)} \pi_T(y \mid x)^\lambda \exp\left(\beta r(x, y)\right),$$

where

$$Z(x) = \sum_y \pi_T(y \mid x)^\lambda \exp\left(\beta r(x, y)\right).$$

Key properties:

- Teacher acts as a soft prior.
- Outcome model reweights sequences toward correctness.
- Single coherent target distribution (not a sum of separate losses).

Distillation Objective

Student policy $\pi_\theta(y \mid x)$.
We choose:

$$\min_\theta \mathrm{KL}\big(\pi^*(\cdot \mid x) \,\|\, \pi_\theta(\cdot \mid x)\big).$$

Expand:

$$\mathrm{KL}(\pi^* \,\|\, \pi_\theta) = \mathbb{E}_{y\sim\pi^*}\big[\log \pi^*(y \mid x) - \log \pi_\theta(y \mid x)\big].$$

Gradient w.r.t. θ:

$$\nabla_\theta \mathrm{KL} = -\mathbb{E}_{y\sim\pi^*}\big[\nabla_\theta \log \pi_\theta(y \mid x)\big].$$

We need to approximate this expectation.

Importance Weights (Student On-Policy)

We sample from the student:

$$y \sim \pi_\theta(\cdot \mid x).$$

For any f:

$$\mathbb{E}_{y\sim\pi^*}[f(y)] = \mathbb{E}_{y\sim\pi_\theta}\left[\frac{\pi^*(y \mid x)}{\pi_\theta(y \mid x)} f(y)\right].$$

$$w_{\text{seq}}(y \mid x) = \frac{\pi^*(y \mid x)}{\pi_\theta(y \mid x)} = \frac{\pi_T(y \mid x)^\lambda \exp(\beta r(x, y))}{\pi_\theta(y \mid x)}.$$

Therefore:

$$\nabla_\theta J(\theta) \mathbb{E}_{y\sim\pi_\theta}\left[w_{\text{seq}}(y \mid x) \nabla_\theta \log \pi_\theta(y \mid x)\right].$$